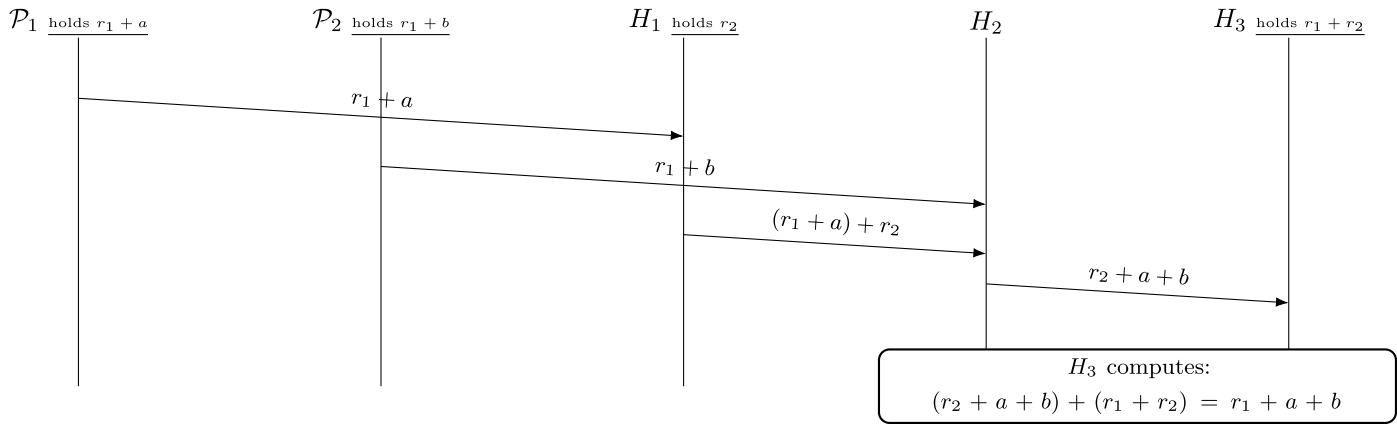
$\mathcal{P}_1$ holds $r_1 + a$
$\mathcal{P}_2$ holds $r_1 + b$
H_1 holds r_2
H_2
H_3 holds $r_1 + r_2$
$r_1 + a$
$r_1 + b$
$(r_1 + a) + r_2$
$r_2 + a + b$
H_3 computes:
$(r_2 + a + b) + (r_1 + r_2) = r_1 + a + b$